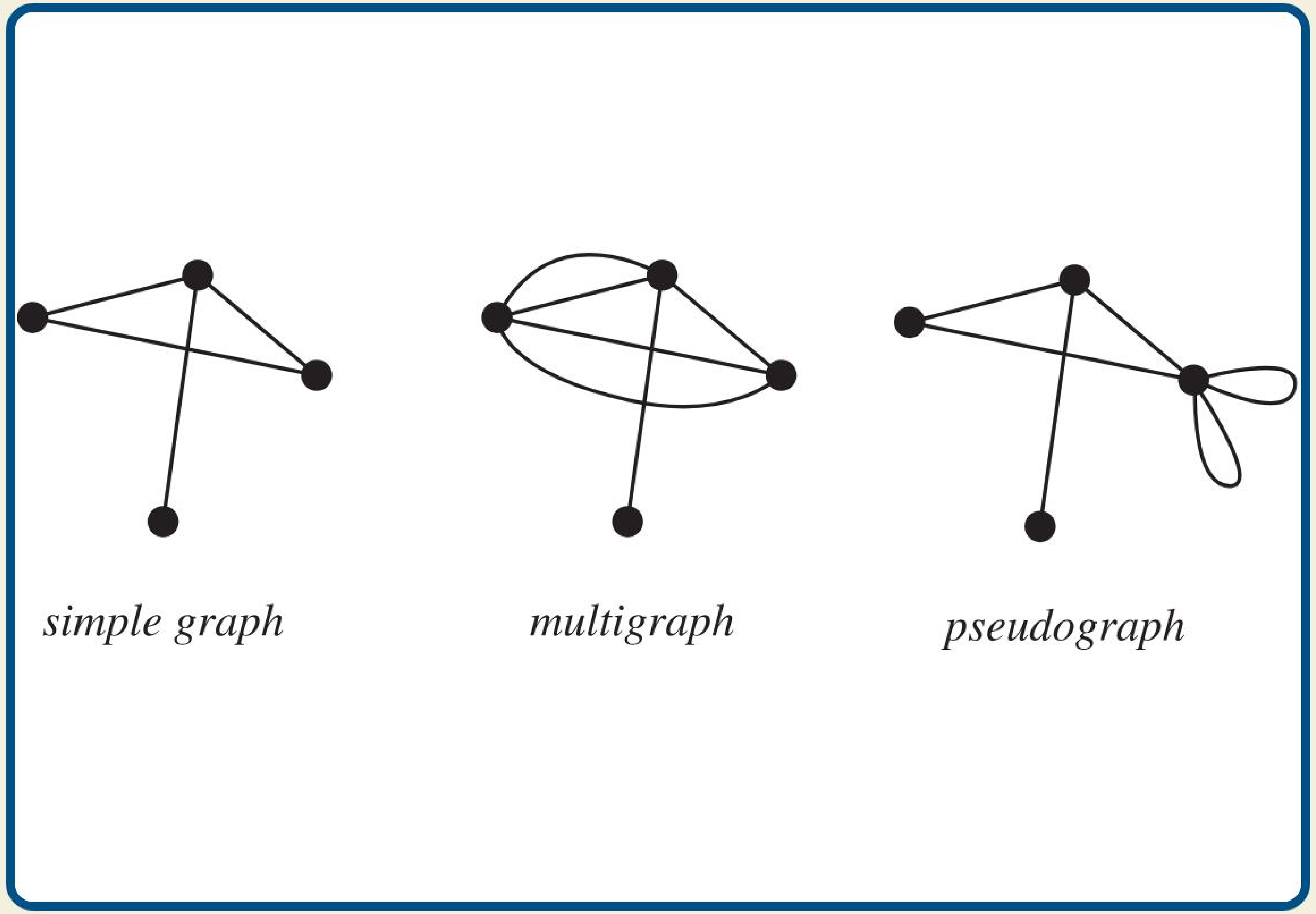
Graphs and Graph Models

Chapter 10 Summary & Key Concepts

Introduction to Graphs

- **Graph G = (V, E):** Consists of a set of vertices (nodes) V and edges E connecting them.
- **Simple Graph:** No loops or multiple edges between the same vertices.
- **Multigraph:** Allows multiple edges between the same pair of vertices.
- **Pseudograph:** Allows loops and multiple edges.
- **Directed Graph (Digraph):** Edges have a direction (ordered pairs).



Graph Terminology

Degree

The degree of a vertex, deg(v), is the number of edges incident to it. A loop contributes 2 to the degree.

Handshaking Thm

For an undirected graph with *m* edges:

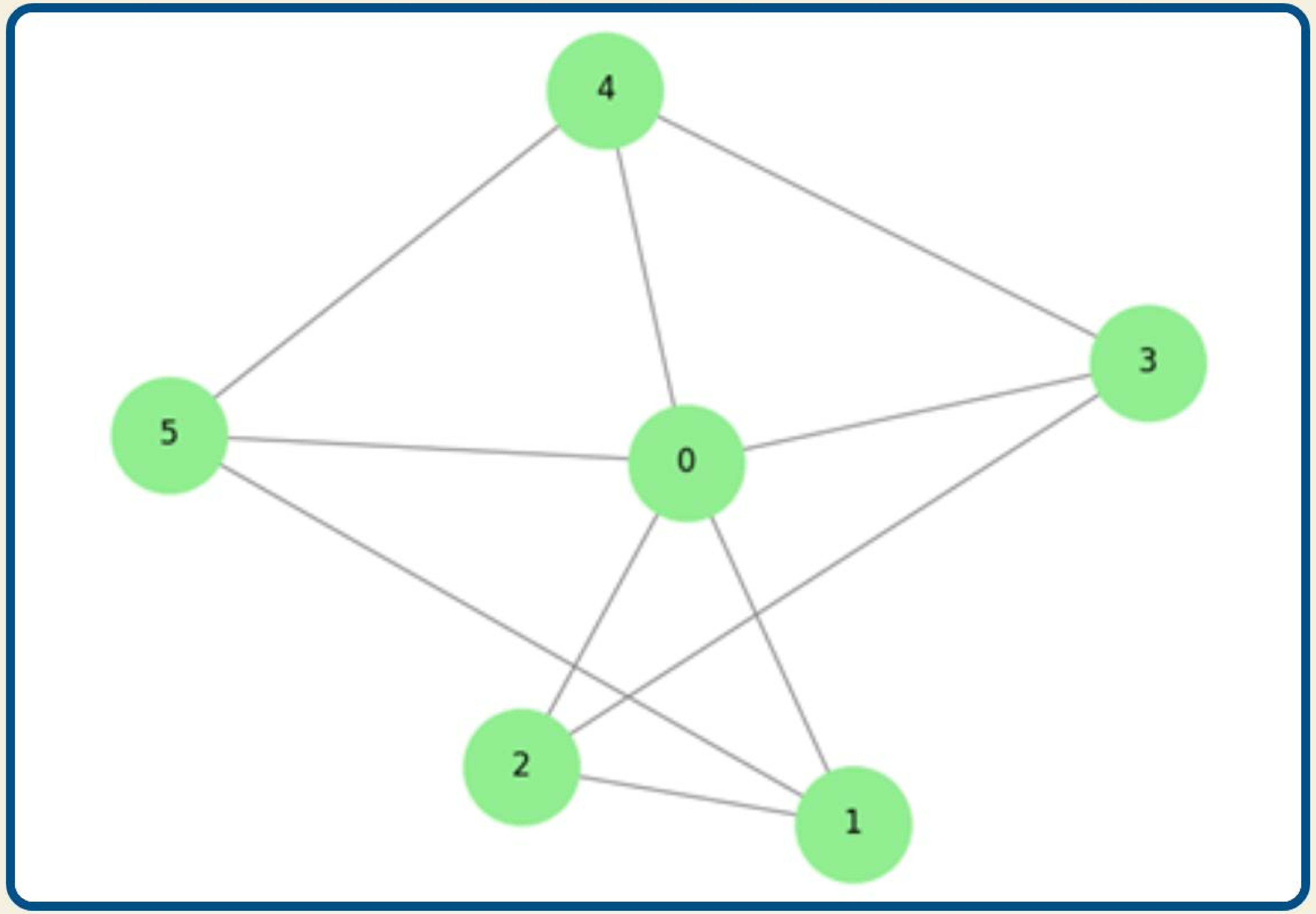
$$2m = \sum_{v \in V} \deg(v)$$

Directed Degree

In-degree: Number of edges ending at v.
Out-degree: Number of edges starting at v.

Special Types of Graphs

- **Complete Graph (K_n):** Every pair of distinct vertices is connected.
- **Cycle (C_n):** A closed loop of n vertices and n edges.
- **Wheel (W_n):** A cycle C_n with an extra vertex connected to all others.
- **n-Cube (Q_n):** Vertices are bit strings of length n; edges connect strings differing by 1 bit.
- **Bipartite:** Vertices partitioned into two sets V_1, V_2 with edges only between sets.



Social Network Models

Friendship Graphs

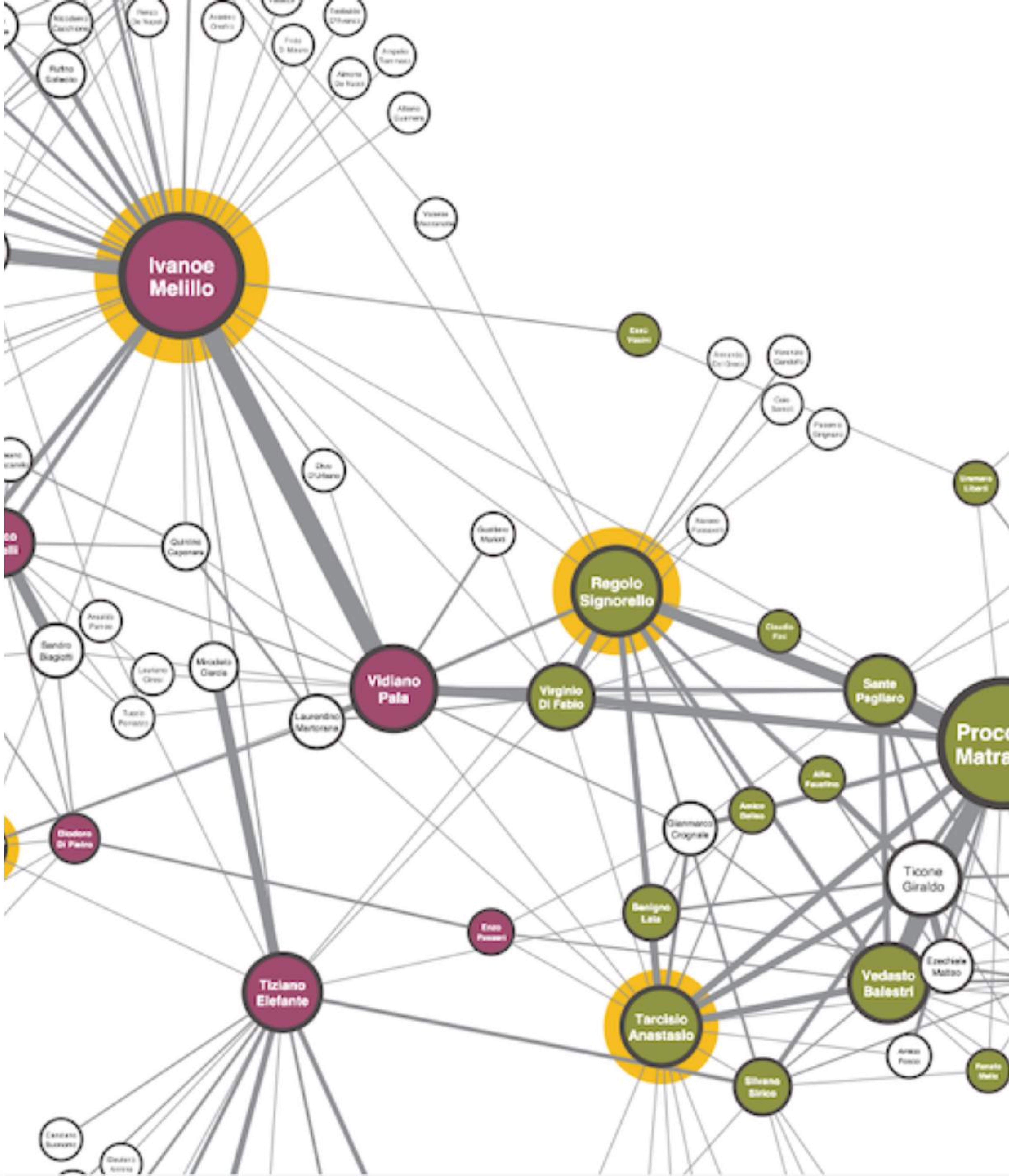
Modeled as undirected graphs where vertices are people and edges represent friendship. Used to analyze cliques and connectivity.

Collaboration Graphs

Vertices represent individuals (e.g., actors, researchers) and edges represent joint work (movies, papers). Examples include the Hollywood graph and academic collaboration graphs.

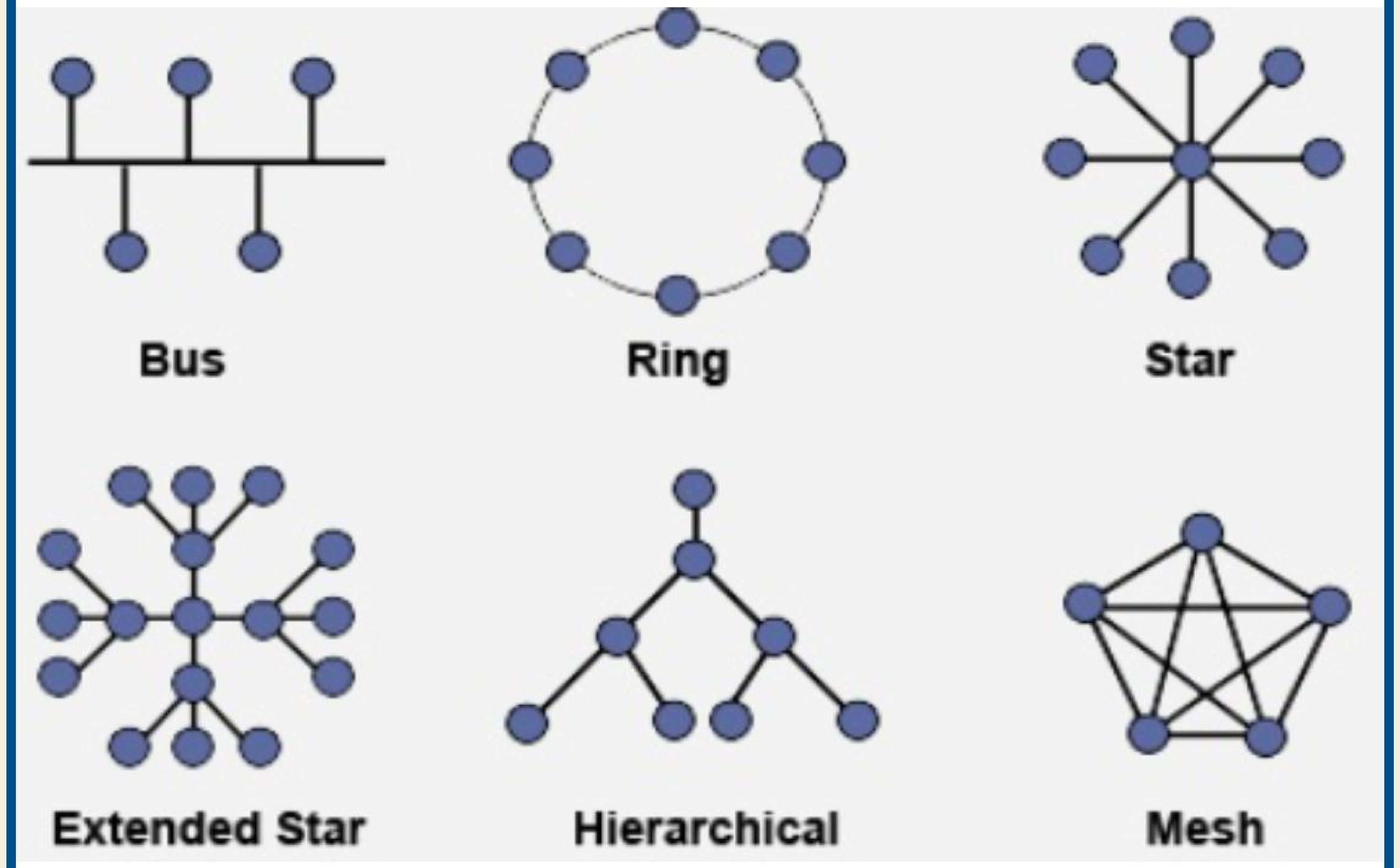
Influence Graphs

Directed graphs modeling how behavior or information spreads from one individual to another.



Infrastructure Models

- **Computer Networks:**
  - *Star:* Central hub connected to all nodes.
  - *Ring:* Each node connected to exactly two others.
  - *Mesh:* Rich connectivity for redundancy.
- **Transportation:** Airports as vertices, flights as directed edges; or Intersections as vertices, roads as edges.
- **Biology:** Protein interaction networks modeling cellular processes.



Representing Graphs

Adjacency Matrix

An n x n matrix **A** where $A_{ij} = 1$ if vertices i and j are adjacent, and 0 otherwise. For multigraphs, the entry counts the number of edges. Symmetric for undirected graphs.

Incidence Matrix

An n x m matrix **M** where rows represent vertices and columns represent edges. $M_{ij} = 1$ if edge j is incident with vertex i. Useful for analyzing edge relationships.

Graph Isomorphism

- **Definition:** Two simple graphs G_1 and G_2 are isomorphic if there is a one-to-one correspondence (bijection) between their vertex sets that preserves adjacency.
- **Invariants:** Properties that must be identical for isomorphic graphs, such as number of vertices, number of edges, and degree sequence.
- **Complexity:** Determining isomorphism is computationally difficult for large graphs. Checking invariants is a necessary but not sufficient condition.

Connectivity

Paths & Circuits

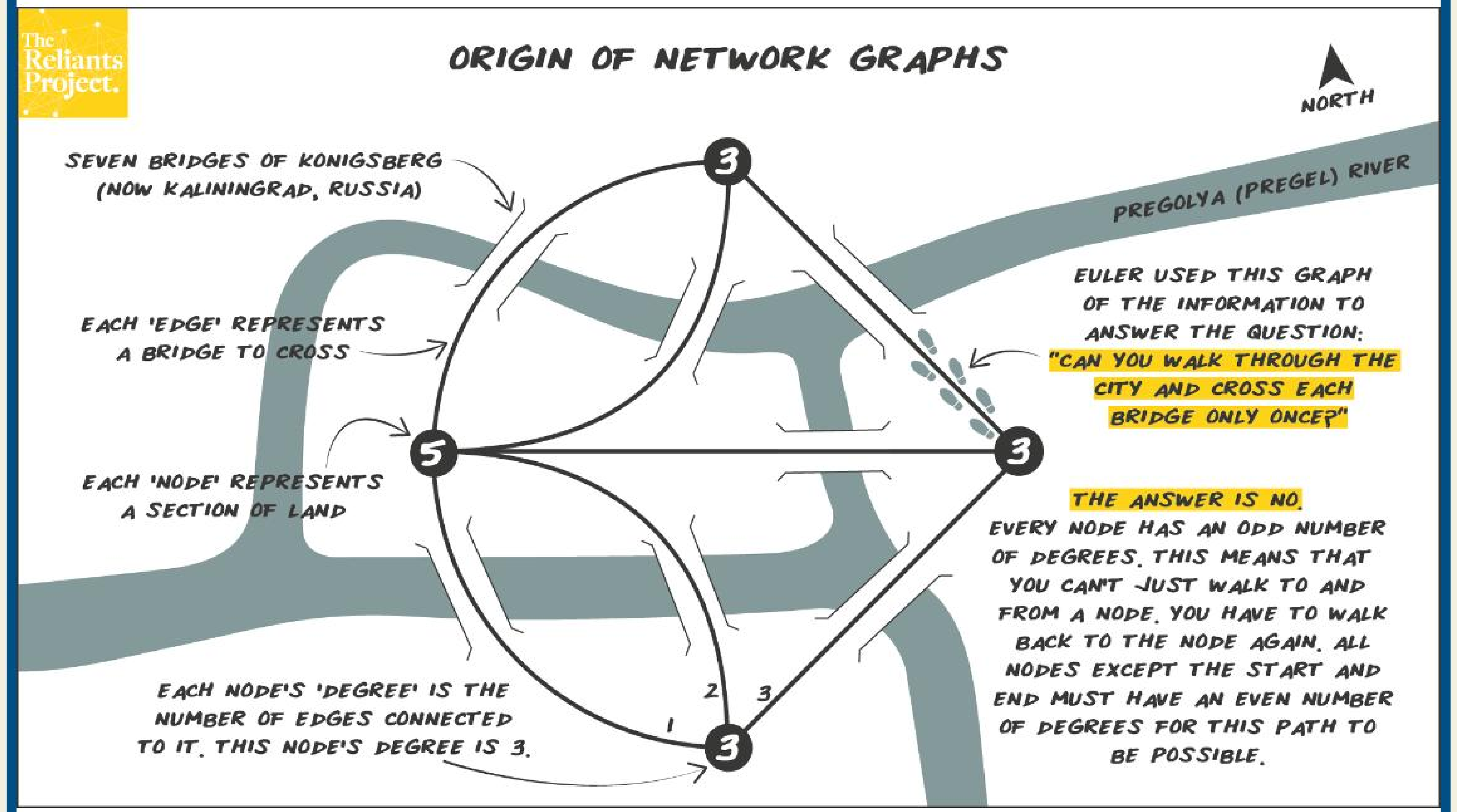
A **path** is a sequence of edges traveling from vertex to vertex. A **circuit** is a path that begins and ends at the same vertex. A graph is **connected** if there is a path between every pair of vertices.

Directed Connectivity

Strongly Connected: A path exists from a to b AND b to a for all vertices.
Weakly Connected: Connected if edge directions are ignored.

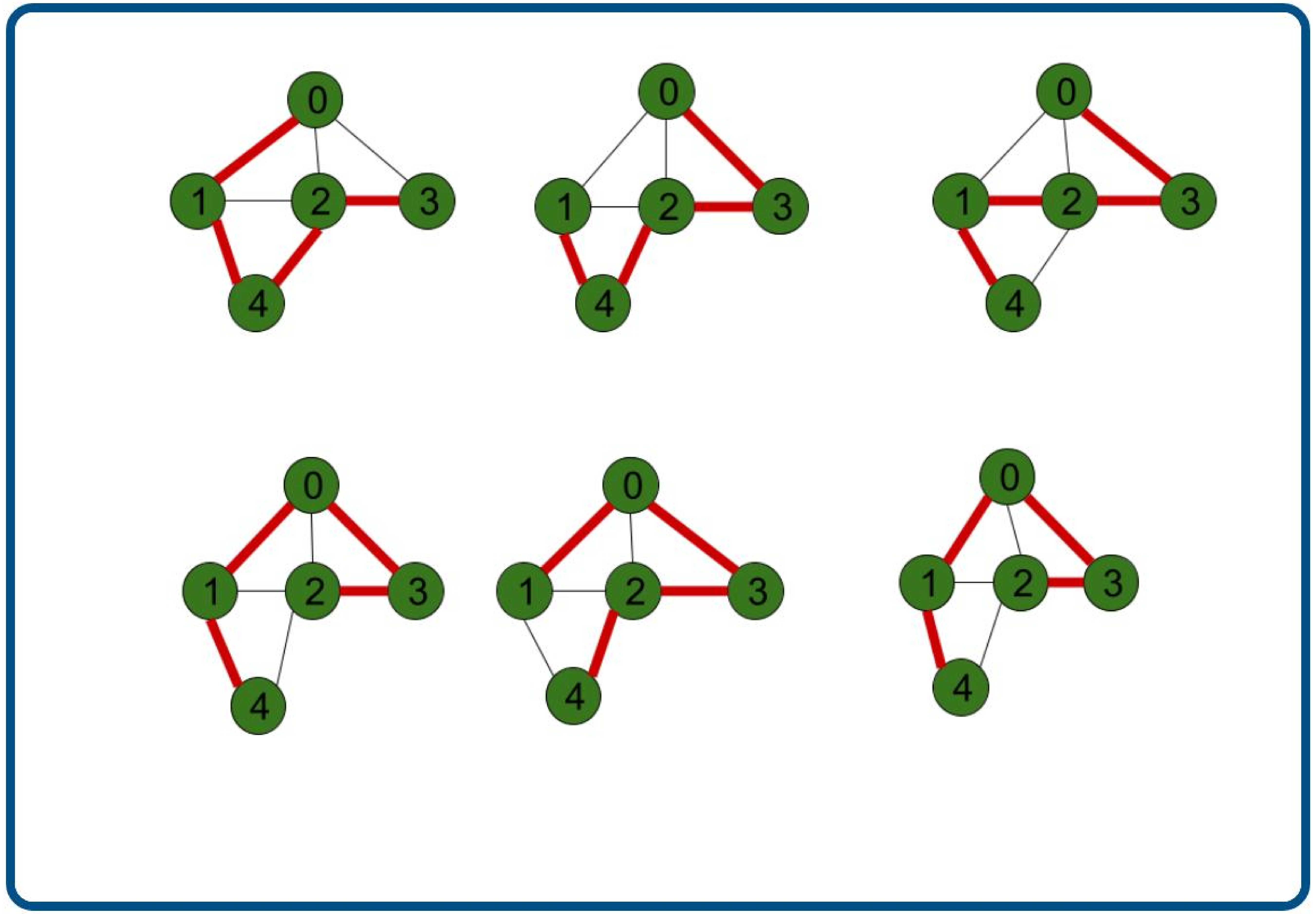
Euler Paths & Circuits

- **Concept:** Traversing every *edge* of a graph exactly once.
- **Euler Circuit:** A circuit containing every edge.
  Condition: Possible iff every vertex has an **even degree**.
- **Euler Path:** A path containing every edge.
  Condition: Possible iff exactly **two vertices** have an odd degree.
- **History:** Solved by Euler regarding the Seven Bridges of Königsberg (pictured).



Hamiltonian Paths & Circuits

- **Concept:** Visiting every *vertex* exactly once.
- **Hamilton Circuit:** Starts and ends at the same vertex, visiting all others once.
- **Theorems:** No simple necessary and sufficient criteria exist. Dirac's and Ore's theorems provide sufficient conditions based on vertex degrees.
- **Application:** The Traveling Salesperson Problem (TSP) seeks the shortest Hamilton circuit in a weighted graph.



Questions?

Thank you for your attention.

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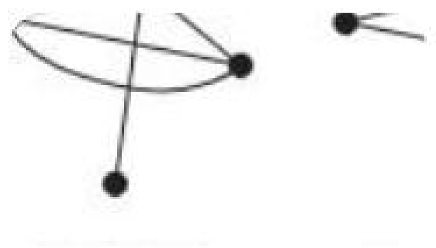
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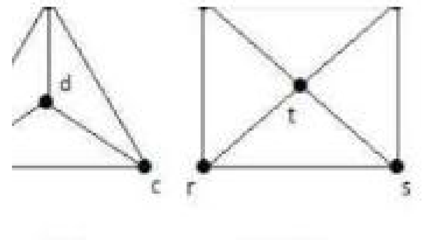
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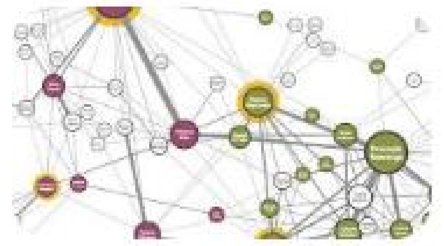
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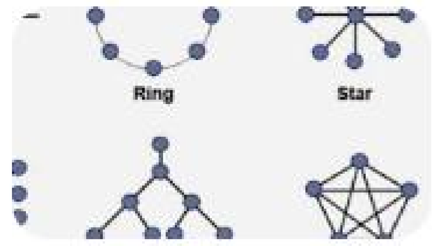

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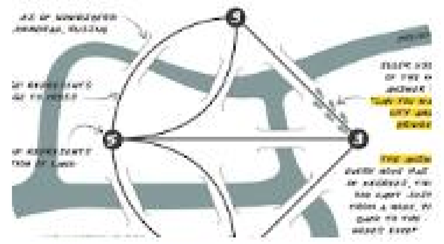
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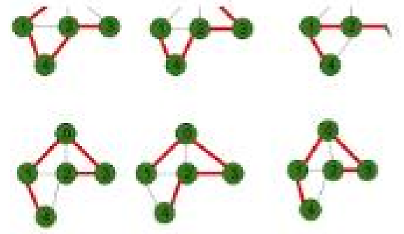
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